LESSON 01 / 05

One tangent can prove a global inequality

If f is differentiable and convex, then for every a and x in its interval,

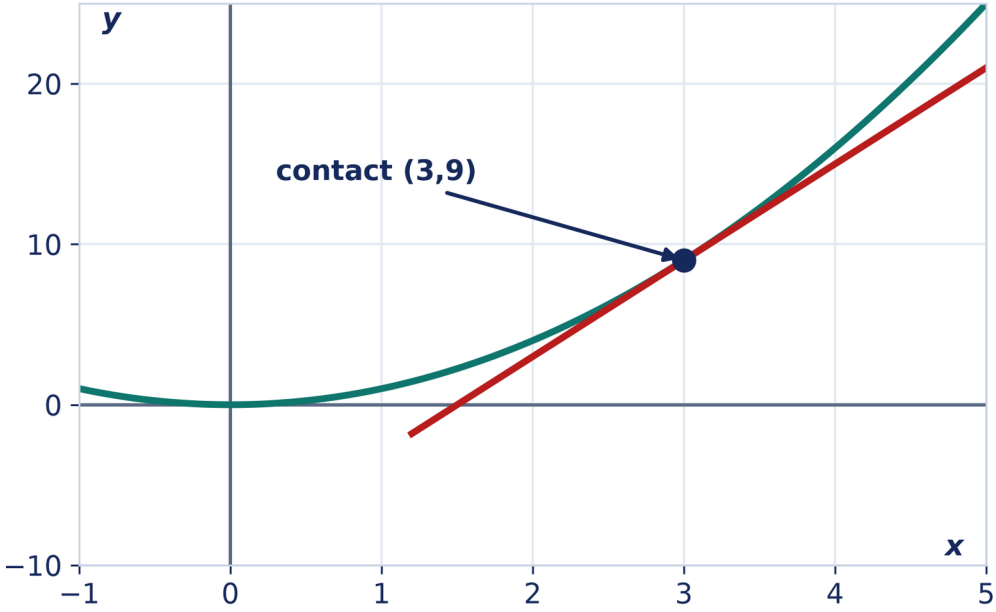
$$f(x) \geq f(a) + f'(a)(x - a).$$

For $f(x) = x^2$ at $a = 3$, the supporting line is

$$y = 9 + 6(x - 3) = 6x - 9.$$

Thus $x^2 \geq 6x - 9$, with equality only at $x = 3$. Independently, the difference is $(x - 3)^2 \geq 0$.

A convex parabola lies above its supporting line



READ THE FIGURE
The red supporting line y=6x-9 touches the teal graph y=x^2 at x=3 and stays below it everywhere. Their difference is (x-3)^2.

Jensen follows by averaging supporting lines

Let $\lambda_i \geq 0$ and $\sum_{i=1}^{n} \lambda_i = 1$, and put $\bar{x} = \sum_{i=1}^{n} \lambda_i x_i$. Apply the supporting-line inequality at $\bar{x}$ to each x_i, multiply by λ_i, and add. The first-order terms cancel because $\sum \lambda_i (x_i - \bar{x}) = 0$. Hence

$$f(\bar{x}) \leq \sum_{i=1}^{n} \lambda_i f(x_i).$$

For strictly convex f, equality requires all the x_i with positive weights to coincide.

WORKED EXAMPLE

Jensen for the exponential

Since $D^2(e^x) = e^x > 0$, *for positive* a, b, c,

$$\frac{e^a + e^b + e^c}{3} \geq e^{(a+b+c)/3},$$

with equality exactly when $a = b = c$. *At* $(1, 2, 3)$, *the left side is approximately* 10.0643 *and the right side is* $e^2 \approx 7.3891$, *so the inequality is strict.*

Convexity of a difference can locate its global minimum

WORKED EXAMPLE

Prove cos x >= 1-x^2/2 for every real x

Let

$$g(x) = \cos x - 1 + \frac{x^2}{2}.$$

Then $g'(x) = x - \sin x$ *and* $g''(x) = 1 - \cos x \geq 0$. *Thus* g' *is increasing. Since* $g'(0) = 0$, *we have* $g' < 0$ *for* $x < 0$ *and* $g' > 0$ *for* $x > 0$. *Therefore* g *decreases up to* 0 *and increases afterwards, so*

$$g(x) \geq g(0) = 0.$$

Equality holds only at $x = 0$. *The argument uses monotonicity of* g', *not an unsupported claim that* $g'' \geq 0$ *alone makes* g *nonnegative.*

LESSON 02 / 05

Young's inequality is weighted AM–GM in exponential form

For $a, b \geq 0$ and conjugate exponents $p, q > 1$ with $1/p + 1/q = 1$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

For positive a, b, weighted AM–GM with weights $1/p$ and $1/q$ gives

$$\frac{a^p}{p} + \frac{b^q}{q} \geq (a^p)^{1/p}(b^q)^{1/q} = ab.$$

Equality holds exactly when $a^p = b^q$, equivalently $a^{p-1} = b$; the boundary equality cases are checked directly.

WORKED EXAMPLE

Young with p=q=2 proves two-variable AM–GM

For $a, b > 0$, apply Young to $\sqrt{a}$ and $\sqrt{b}$:

$$\sqrt{ab} \leq \frac{(\sqrt{a})^2 + (\sqrt{b})^2}{2} = \frac{a+b}{2}.$$

Equality holds exactly when $\sqrt{a} = \sqrt{b}$, hence when $a = b$.

AM–GM for three positive numbers from concavity

Since $(\ln x)'' = -1/x^2 < 0$, Jensen for the concave logarithm reverses the inequality:

$$\frac{\ln x + \ln y + \ln z}{3} \leq \ln\left(\frac{x+y+z}{3}\right).$$

Exponentiating gives

$$\sqrt[3]{xyz} \leq \frac{x+y+z}{3},$$

with equality exactly when $x = y = z$. This proves Q36. The same concave-log Jensen argument with weights $1/p$ and $1/q$ proves the weighted AM–GM step used above.

Cauchy–Schwarz starts with a direct square

For all real a, b,

$$a^2 + b^2 - 2ab = (a-b)^2 \geq 0,$$

so $a^2 + b^2 \geq 2ab$, with equality exactly when $a = b$. The source's alternative expression containing $\sqrt{a^2b^2}$ equals $|ab|$, not ab; it is invalid when $ab < 0$. The displayed square is valid for all real numbers.

Summing the two-variable inequality with suitable normalisation leads to Cauchy–Schwarz. In Engel form, for real a_i and positive x_i,

$$\left(\sum_{i=1}^{n} a_i\right)^2 = \left(\sum_{i=1}^{n} \frac{a_i}{\sqrt{x_i}}\sqrt{x_i}\right)^2 \le \left(\sum_{i=1}^{n} \frac{a_i^2}{x_i}\right)\left(\sum_{i=1}^{n} x_i\right).$$

Thus $\sum a_i^2/x_i \ge (\sum a_i)^2/\sum x_i$.

The inequality family has distinct hypotheses and equality cases

Result	Main hypotheses	Equality
Young	1/p+1/q=1	a^p=b^q
2-variable AM-GM	a,b>0	a=b
3-variable AM-GM	x,y,z>0	x=y=z
Square bound	a,b real	a=b
Cauchy Engel	x_i>0	a_i proportional to x_i

READ THE FIGURE

Use the signed square for all real a,b. Use positive variables when logarithms, geometric means or Engel denominators require them.

LESSON 03 / 05

Differentiate the family with respect to its parameter

For $a > 0$ and $m \neq 0$, consider

$$F(x, y, m) = y - mx - \frac{a}{m} = 0.$$

An envelope point satisfies both $F = 0$ and $\partial F/\partial m = 0$:

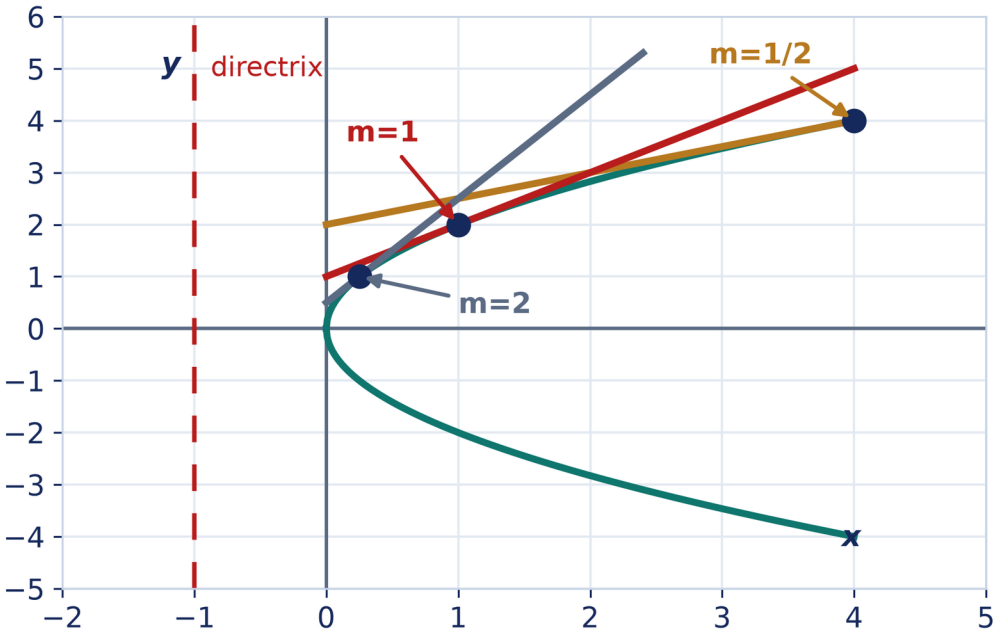
$$-x + \frac{a}{m^2} = 0 \quad \Longrightarrow \quad x = \frac{a}{m^2}.$$

Substitution into the line gives $y = 2a/m$. Therefore the contact point is

$$\left(\frac{a}{m^2}, \frac{2a}{m}\right),$$

and eliminating m yields $y^2 = 4ax$. Thus every line $y = mx + a/m$ touches this parabola at the stated point.

A family of lines envelops the parabola y^2=4x



READ THE FIGURE
For a=1, the gold, red and slate lines have m=1/2,1,2 and touch at the marked points. The red dashed line x=-1 is the directrix used in the perpendicular-tangent theorem.

Perpendicular tangents meet on the directrix

A tangent perpendicular to $y = mx + a/m$ has slope $-1/m$. Replacing m by $-1/m$ in the family gives

$$y = -\frac{x}{m} - am.$$

At the intersection of the two tangents,

$$mx + \frac{a}{m} = -\frac{x}{m} - am,$$

so

$$(m + \tfrac{1}{m})(x + a) = 0.$$

Since $m + 1/m \neq 0$ for real m, the intersection has $x = -a$, the directrix of $y^2 = 4ax$.

COMMON TRAP

Envelope differentiation uses the parameter, not x or y

The equation $\partial F/\partial m = 0$ is taken while x and y are held fixed. It supplies the second equation needed to locate the envelope and its contact point.

LESSON 04 / 05

Perpendicular ellipse tangents determine a concentric circle

For the ellipse $x^2/a^2 + y^2/b^2 = 1$, a nonvertical tangent of slope m can be written

$$y = mx \pm \sqrt{a^2m^2 + b^2}.$$

If it passes through $P = (x, y)$, rearranging and squaring gives

$$(x^2 - a^2)m^2 - 2xym + (y^2 - b^2) = 0.$$

The two tangent slopes from P are perpendicular exactly when their product is -1. By the product of roots,

$$\frac{y^2 - b^2}{x^2 - a^2} = -1 \quad \Longrightarrow \quad x^2 + y^2 = a^2 + b^2.$$

This is the **director circle**. Vertical or horizontal tangent cases satisfy the same locus directly.

WORKED EXAMPLE

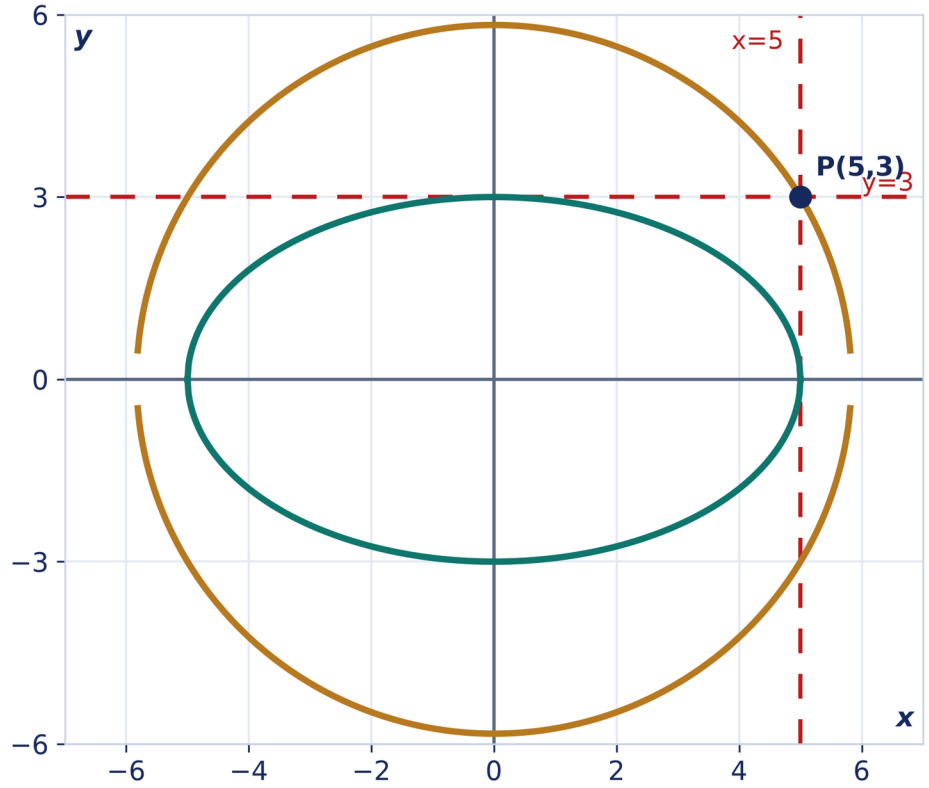
The 5-3 ellipse and the circle special case

For $x^2/25 + y^2/9 = 1$, perpendicular tangents meet at P with

$$|OP|^2 = 25 + 9 = 34, \qquad |OP| = \sqrt{34} \approx 5.831.$$

When $a = b = R$, the ellipse is a circle and the director circle has radius $\sqrt{2}R$. Thus perpendicular tangents to $x^2 + y^2 = R^2$ meet at distance $\sqrt{2}R$ from the centre.

Perpendicular tangents to the ellipse meet on its director circle



READ THE FIGURE
The teal ellipse has perpendicular tangents x=5 and y=3 at (5,0) and (0,3). They meet at P=(5,3), which lies on the gold director circle x^2+y^2=34.

Lagrange multipliers encode a constrained stationary point

To extremise f subject to $g=0$, a regular interior extremum satisfies $\nabla f=\lambda\nabla g$. For

$$f=x^2+y^2+z^2, \qquad g=x+y+z-3=0,$$

this gives $2x=2y=2z=\lambda$. Hence $x=y=z$, and the constraint gives $x=y=z=1$. The candidate value is 3.

A global proof is immediate:

$$x^2+y^2+z^2-3=(x-1)^2+(y-1)^2+(z-1)^2\geq 0$$

when $x+y+z=3$. Thus the minimum is 3 at $(1,1,1)$, not merely a stationary candidate.

COMMON TRAP
Multipliers find candidates; a separate argument establishes global extrema
Check regularity, boundaries and attainment. Here the completed squares supply the missing global comparison in one line.

LESSON 05 / 05

Symmetry suggests equal variables; monotonicity in the product proves it

For positive x, y with fixed sum t, put $s = xy$. Then

$$0 < s \le \frac{t^2}{4},$$

with equality at $x = y = t/2$. A symmetric expression rewritten in terms of t and s is optimised by checking whether it increases or decreases in s.

WORKED EXAMPLE

Two minima with x+y=10

Since $s = xy \le 25$,

$$\frac{1}{x} + \frac{1}{y} = \frac{10}{s} \ge \frac{10}{25} = \frac{2}{5},$$

and

$$x^2 + y^2 = 100 - 2s \ge 100 - 50 = 50.$$

Both minima occur at $x = y = 5$.

For positive x, y, z with $x + y + z = 1$, AM–GM gives

$$xyz \le \left(\frac{x+y+z}{3}\right)^3 = \frac{1}{27},$$

with equality at $x = y = z = 1/3$. A one-variable alternative fixes x and uses $yz \le (1-x)^2/4$, then maximises $x(1-x)^2/4$ by differentiation.

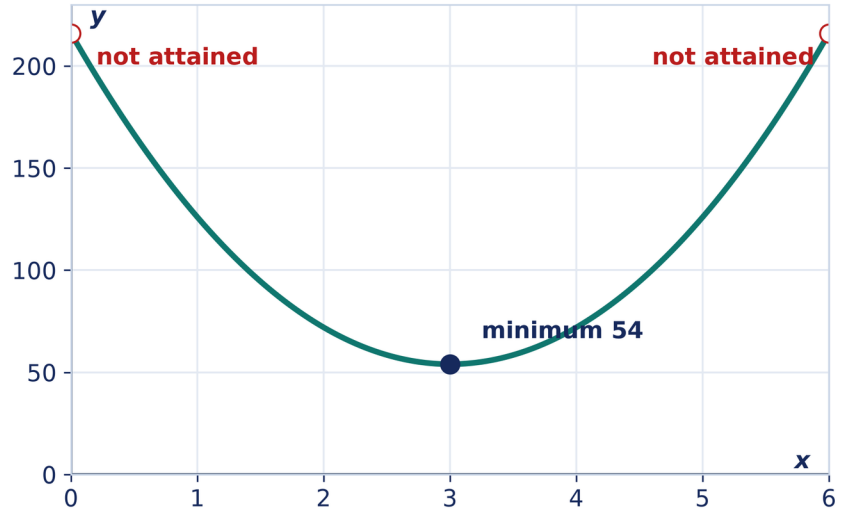
The symmetric cubic is a counterexample to blind equalising

For positive x, y with $x + y = 6$,

$$x^3 + y^3 = 216 - 18xy = 18(x-3)^2 + 54.$$

Since $xy \le 9$, the attained minimum is 54 at $x = y = 3$. But as $x \to 0^+$ and $y \to 6$, the value approaches 216 from below. Therefore the **supremum is 216 and no maximum is attained** on the positive domain.

The symmetric cubic has an interior minimum but no attained maximum



READ THE FIGURE
The graph shows x^3+y^3 after substituting y=6-x. Hollow red endpoints mark limits outside the positive domain; the interior point (3,54) is the attained minimum.

As a numerical check, at $(x,y)=(0.1,5.9)$ the value is $0.001+205.379=205.38$, close to but below 216. This corrects the companion solution's printed check value 205.8.

Equal variables minimise a sum of squares under a plane constraint

For positive x,y,z with $x+y+z=1$,

$$\left(x-\tfrac{1}{3}\right)^2+\left(y-\tfrac{1}{3}\right)^2+\left(z-\tfrac{1}{3}\right)^2=x^2+y^2+z^2-\tfrac{1}{3}.$$

Hence

$$x^2+y^2+z^2\geq\tfrac{1}{3},$$

with equality exactly when $x=y=z=1/3$. The same completed-square argument proves the Lagrange example in Extension IV.

COMMON TRAP
Always identify whether the objective increases or decreases in s
Equal variables maximise xy and minimise x^2+y^2 for a fixed positive sum, but they minimise rather than maximise x^3+y^3. Symmetry alone does not choose the type of extremum.